Mental Imagery Beyond Vividness: A Methodological Proposal for Investigating Temporal Stability and Incompletion-Based Perturbations

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Abstract

Mental imagery research has primarily focused on vividness, controllability, and individual differences, whereas the temporal persistence of internally generated representations has received comparatively less attention. Existing evidence suggests that imagery vividness declines from adolescence to adulthood, yet it remains unclear whether interventions targeting closure dynamics and automatic completion processes can influence the vividness and temporal stability of mental imagery.

The present work proposes an exploratory protocol designed to investigate whether brief incompletion-based perturbations may influence two related dimensions of mental imagery. We introduce a set of micro-interventions designed to transiently interrupt automatic closure processes without explicitly training imagery ability, alongside operational descriptions of common sources of drift, including monitoring, optimization, closure, and ritualization. Rather than treating these failure modes solely as exclusion criteria, we examine whether making them explicit influences task outcomes.

Participants complete repeated sessions in which a brief intervention precedes the generation of a simple mental image. Primary outcomes are subjective imagery vividness and perceived temporal stability, defined as the ease with which an image remains present over time without continuous reconstruction or effortful maintenance. Secondary measures include perceived effort, spontaneous transformation, reconstruction frequency, and image disappearance.

We explore whether incompletion-based perturbations are associated with changes in imagery vividness and temporal stability. We additionally explore whether explicit awareness of common failure modes interacts with performance in potentially non-linear ways. More broadly, this paper proposes that transient modifications of closure dynamics may represent an underexplored determinant of mental imagery phenomenology.

1 Introduction

Mental imagery refers to the ability to generate internally driven sensory-like representations in the absence of direct sensory input and contributes to memory, future simulation, planning, creativity, and subjective experience (Pearson et al., 2015). However, imagery abilities vary substantially across individuals, spanning a continuum from aphantasia to hyperphantasia, suggesting large interindividual differences in how internally generated representations are experienced and maintained (Keogh & Pearson, 2018).

Among the dimensions most frequently investigated, imagery vividness has received particular attention. Vividness generally refers to the subjective clarity, perceptual richness, or intensity of internally generated representations and has traditionally been assessed using self-report instruments such as the Vividness of Visual Imagery Questionnaire, VVIQ (Marks, 1973). This approach has generated substantial evidence for large variability in imagery phenomenology across individuals.

Recent evidence further suggests that imagery vividness changes across the lifespan. Using a large cross-sectional sample from adolescence to middle adulthood, previous work reported a progressive reduction in self-reported imagery vividness, accompanied by a decrease in highly vivid imagery profiles and an increase in lower-vividness imagery across maturation and aging. These findings suggest that developmental processes substantially influence internally generated representations across the lifespan.

Despite this literature, research has focused predominantly on imagery vividness, imagery manipulation, and controllability, whereas the temporal persistence of internally generated representations has received comparatively less direct attention. Processes related to imagery maintenance have often been examined indirectly through visual working memory frameworks, where maintaining internally generated information across time constitutes a central operation (Baddeley & Hitch, 1974). However, maintenance has usually been treated as a cognitive process rather than as a phenomenological variable in itself.

This distinction may be important because imagery vividness and imagery stability are not necessarily equivalent constructs. A mental representation may be highly vivid yet disappear rapidly, whereas another may remain present over time while exhibiting relatively low perceptual richness. We therefore define temporal stability as the subjective ease with which an internally generated representation remains present over time without continuous reconstruction, rehearsal, or effortful maintenance.

The present work focuses simultaneously on these two dimensions of imagery phenomenology: vividness and temporal stability. We investigate whether brief incompletion-based perturbations influence these dimensions by transiently introducing unresolved cognitive states prior to image generation. These interventions are accompanied by explicit descriptions of common sources of operational drift, including monitoring, optimization, closure pressure, and proceduralization. In the present framework, operational drift is treated as a primary methodological concern rather than a secondary source of noise, because repeated exposure may progressively alter the cognitive organization that the protocol aims to examine. The present work should be interpreted primarily as a methodological proposal rather than a confirmatory investigation. The objective is not to establish effectiveness, but to examine whether these constructs, measurements, and operational procedures provide a useful framework for studying imagery persistence and phenomenology.

Importantly, the present work does not assume gradual learning or monotonic improvement. Instead, repeated testing is conceptualized as probabilistic sampling in which interventions may transiently alter the likelihood of producing imagery states characterized by increased vividness or temporal stability. Under this framework, improvement may emerge discontinuously through isolated high-imagery sessions rather than progressive linear gains.

Accordingly, the primary objective of this work is to examine whether incompletion-based perturbations are associated with differences in the subjective vividness and temporal stability of mental imagery during repeated low-frequency testing across seven days.

2 Method

Methods

Participants

Participants complete a seven-day repeated-measures protocol designed to assess whether incompletion-based perturbations alter imagery vividness and temporal stability. Testing frequency is intentionally low, one session per morning, to reduce rapid proceduralization and minimize optimization across repeated attempts.

Procedure

Participants complete one session each morning for seven consecutive days. Each session follows the same structure:

Step 1: Brief review of the assigned intervention and associated failure points.

Step 2: Execution of the intervention for approximately 20-60 seconds.

Step 3: Generation of a simple mental image. Participants are instructed to generate a neutral image such as an object, place, face, geometric form, or scene.

Step 4: Observation period. Participants allow the image to remain present for a short period without explicit instructions to maximize effort, improve performance, or maintain concentration.

Step 5: Immediate self-report measures.

Testing Environment

Sessions are performed within a standardized virtual environment consisting of older futuristic first-person games played at low difficulty. The environment should allow continuous movement, rapid perceptual dynamics, low narrative demands, and frequent transient stimuli while minimizing performance pressure. When possible, the HUD is reduced or disabled to limit score tracking, objective monitoring, and other evaluative cues.

Interventions

Interventions consist of brief incompletion-based perturbations designed to transiently introduce unresolved cognitive states before image generation. Examples include internally asking incomplete questions without answering them, briefly initiating unfinished cognitive sequences, or similar operations intended to reduce immediate closure tendencies.

To reduce operational drift, interventions are accompanied by descriptions of common sources of drift including monitoring, optimization, closure pressure, question collection, and ritualization.

Primary Measures

Participants rate the following primary outcomes after each session:

Ease of maintaining the image over time 0 = disappeared immediately 10 = remained present easily with little effort

Imagery vividness 0 = absent or extremely weak 10 = extremely vivid or perceptually rich

Secondary Measures

Participants additionally rate:

Effort required to maintain the image Image transformation frequency Reconstruction frequency Perceived disappearance rate

Additional effects checklist. Participants may select any additional effects noticed after the trial.

- ☐ Chest pleasure
- ☐ Fatigue or mental tiredness
- ☐ Increased calmness
- ☐ Nervousness or tension
- ☐ Increased distractibility
- ☐ Image became unusually persistent
- ☐ Other: ____________________________

Analytical Framework

The framework does not assume monotonic improvement across repeated sessions. Instead, repeated trials are conceptualized as probabilistic sampling opportunities in which interventions may transiently alter the likelihood of producing imagery states characterized by increased vividness or temporal stability.

Consequently, analyses focus primarily on:

* frequency of high-imagery sessions, * within-subject variability, * probability of unusually stable or vivid sessions, * discontinuous or threshold-like response patterns,

rather than linear improvement across days.

3 Discussion

The present work introduced an exploratory framework designed to examine whether incompletion-based perturbations influence imagery vividness and temporal stability. Rather than assuming that repeated exposure produces gradual improvements through training, the method was designed around a probabilistic framework in which interventions may transiently modify the likelihood of entering imagery states characterized by greater persistence or perceptual richness.

If imagery vividness and temporal stability increase following incompletion-based perturbations, this would suggest that relatively small modifications in cognitive organization prior to image generation can alter subsequent imagery phenomenology. Such findings would support the view that imagery quality depends not only on stable imagery ability itself but also on transient cognitive configurations preceding imagery generation (Pearson et al., 2015).

An important implication concerns the distinction between vividness and stability. Much of the existing literature has emphasized vividness as the dominant phenomenological dimension of imagery, typically operationalized using instruments such as the VVIQ (Marks, 1973). However, the present framework assumes that vividness and persistence may be partially separable dimensions. A mental representation may be highly vivid but rapidly disappear, whereas another may remain present over time while exhibiting relatively limited perceptual richness. Differential intervention effects across these dimensions would support the existence of partially independent phenomenological components of imagery.

The present work also differs from traditional imagery frameworks by emphasizing temporal persistence as an outcome variable. Processes related to imagery maintenance have often been investigated indirectly through working memory frameworks, where maintenance functions primarily as a cognitive resource problem rather than as a phenomenological variable in itself (Baddeley & Hitch, 1974; Baddeley, 2000). The present findings therefore raise the possibility that subjective imagery persistence may represent an understudied dimension of imagery research.

The probabilistic structure of the protocol generates predictions that differ from conventional learning models. Rather than expecting monotonic improvement across repeated sessions, the present framework predicts substantial within-subject variability, isolated high-imagery sessions, and discontinuous response patterns. Such dynamics are more consistent with state-dependent or threshold-like interpretations in which temporary configurations alter the probability of entering favorable representational states rather than progressively strengthening a stable skill (Kelso, 1995; van der Maas et al., 2006).

Several limitations should also be considered. First, the method relies heavily on self-report measures. While self-report remains central in imagery research, introspective access may itself alter cognitive processing, particularly under repeated evaluation conditions (Schooler, 2002). Repeated self-evaluation may therefore modify imagery experience across sessions and partially influence observed trajectories.

Second, participants may not clearly distinguish between vividness, persistence, reconstruction effort, and image disappearance. These constructs may overlap phenomenologically, producing measurement noise and interpretative ambiguity. Future studies may benefit from more refined phenomenological methods or behavioral proxies for imagery maintenance.

Third, the interventions themselves are intentionally heterogeneous and may recruit multiple processes simultaneously. Monitoring reduction, interruption of closure processes, expectancy effects, attentional changes, or working memory dynamics may all contribute to observed effects, limiting strong mechanistic interpretation.

Finally, individual differences in imagery ability remain substantial across populations, spanning conditions ranging from aphantasia to hyperphantasia (Zeman et al., 2020). Consequently, intervention effects may interact strongly with baseline imagery characteristics.

The present work proposes that imagery phenomenology may depend not only on representational capacity but also on transient cognitive configurations that precede imagery formation. If replicated, these findings would suggest that temporal stability itself constitutes an important and comparatively understudied dimension of mental imagery research.

Environmental context may substantially shape intervention effects. Older futuristic first-person games may provide useful testing environments due to their combination of rapid perceptual dynamics, reduced narrative structure, continuous sensorimotor engagement, and rich transient stimuli. Pilot observations suggest that such environments may facilitate implementation of the present interventions, although direct comparisons across contexts remain necessary. However, reducing HUD information and difficulty may also limit generalizability to more demanding gameplay contexts.

More broadly, if monitoring and closure processes influence imagery persistence, similar mechanisms may also contribute to variability in other domains involving internally generated cognition, including creativity and positive affect. Excessive monitoring may not only interfere with imagery maintenance but could also alter access to cognitive states that depend on reduced evaluative interruption.

4 Conclusion

The present work proposes an exploratory framework for investigating whether brief incompletion-based perturbations influence two dimensions of mental imagery, vividness and temporal stability. While imagery research has historically emphasized vividness, controllability, and individual differences, comparatively less attention has been devoted to the subjective ease with which internally generated representations remain present over time.

The paper adopts a probabilistic framework in which repeated attempts are conceptualized not as training opportunities but as repeated sampling occasions that may transiently alter imagery outcomes. Under this perspective, variability, discontinuity, and isolated high-imagery sessions may be as informative as average performance changes.

More broadly, this work introduces temporal stability as a candidate dimension of imagery phenomenology and explores whether transient cognitive states preceding image generation contribute to subsequent imagery experience. Given the exploratory nature of the present approach, the primary contribution of this work is not to establish effectiveness, but to examine whether these constructs and methods provide useful directions for future investigation of mental imagery dynamics.

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5 References

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6 Appendix

6.1 Operational Drift Guide

These sections are intended as operational aids rather than performance criteria. Participants may briefly consult them before or between attempts to identify whether the exercise gradually shifted toward monitoring, optimization, excessive control, or procedural repetition.

6.2 Operational Drift Guide: Open Question

Ask yourself a question internally, but do not answer it.

Recruit answer. Many questions automatically recruit answer generation. Even simple questions such as “What if I turned left?” may trigger prediction, simulation, or pros/cons evaluation.

Meta-monitoring. You may begin checking whether you successfully avoided answering. The exercise can then become a performance evaluation task.

Question collection. The brain may start generating multiple questions because the exercise itself becomes a search process, for example: "find a good incomplete question."

Repetition ritualization. Repeated use of the same question format may create procedural routines. For example, always using spatial questions or similar structures.

Open loop persistence. Some questions may remain active and continue occupying working memory after the exercise moment ends.

Goal contamination. The question may become instrumentalized, for example: "this question will lower control" or "this question should improve imagery."

6.3 Operational Drift Guide: Unfinished Sentence

Do not finish a sentence you are thinking about.

Completion monitoring. The brain may immediately ask: "Did I leave it unfinished?" "Did I accidentally finish it?" or "How long should it remain incomplete?" Incompleteness itself may become monitored.

Reactivation. Unfinished cognitive objects often attract reactivation. Instead of disappearing, the sentence may persist through repeated retrieval attempts such as: "What was I going to say?"

Deliberate sentence construction. A procedural sequence may emerge: invent sentence, interrupt sentence, maintain incompleteness. The exercise then becomes highly structured.

Internal language overload. Because the exercise relies on language, it may recruit semantic processing, self-talk, rehearsal, or internal narration. Linguistic exercises may stabilize more easily than short sensorimotor perturbations.

Turning uncertainty into content. Incomplete sentences may drift toward planning or simulation loops, for example: "What if..." or "Maybe I should..."

Completion pressure. Unfinished structures often recruit closure tendencies. The urge to complete the sentence may itself become salient.

Repetition effects. Repeated use may create recurring templates such as "maybe...", "what if...", or "I should...", making the exercise increasingly formulaic.

Meta-monitoring. A common loop is checking whether the sentence remained sufficiently unresolved, transforming incompletion into evaluation.

Narrative generation. The unfinished sentence may gradually acquire meaning or future orientation, creating stable narrative objects.

Hidden task creation. The exercise may silently become "practice incompletion." Once it acquires a training objective, monitoring pressure may increase.

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